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Non-traditional HIIT-style ROTC training elicits positive bone quality and performance adaptations

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ABSTRACT

Military personnel experience elevated bone injury incidence, partly due to arduous and repetitive training. Non-traditional High-Intensity Interval Training-style (HIIT) may benefit pre-enlisted Reserve Officer Training Corps (ROTC) cadet's musculoskeletal health and performance prior to military service. This study investigated 16 ROTC ($n = 12$ males; $n = 4$ females) and 15 physically active sex-, age-, and body mass-matched Controls' musculoskeletal health and performance from November to April. Total body, lumbar spine, and dual-hip dual-energy X-ray absorptiometry scans and 4%, 38%, 66% tibial peripheral quantitative computed tomography scans, blood draws (serum sclerostin and parathyroid hormone), and maximal muscle strength and aerobic capacity testing were completed. From November to April, ROTC improved bone density (DXA) of the dominant total hip and greater trochanter and non-dominant greater trochanter and 38% and 66% tibial total volumetric and cortical bone density (pQCT) similarly or more than Controls (all $p \leq 0.049$). From November to April, ROTC also improved bench and leg press, and peak aerobic capacity (all $p \leq 0.013$). From November to January, serum sclerostin increased ($p \leq 0.007$) and remained elevated through April, while parathyroid hormone was unchanged. HIIT-style training induced positive musculoskeletal adaptations, suggesting it may be an excellent pre-service training modality for this injury prone group.

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KEYWORDS

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Introduction

The dynamic nature of bone turnover induced by exercise or physical training consists of skeletal and endocrine responses to internal and external loading stimuli (Proff & Römer, 2009; Smith et al., 2021) and is often assessed using skeletal imaging technologies and serum biomarkers. Specific imaging systems such as dual-energy X-ray absorptiometry (DXA) and peripheral quantitative computed tomography (pQCT) measure bone density, content, and geometric characteristics at a macro-level. DXA also can provide information regarding bone-free lean body mass and fat mass, offering an encompassing assessment of bone health and body composition for whole body and segmental regions (Varela & Jolette, 2018). Bone geometric properties can be measured by pQCT, providing estimations of bones' morphology and structural integrity (Varela & Jolette, 2018). At a micro-level, many biomarkers are used to understand rates of bone turnover; these are called bone turnover markers (BTMs). For instance, cross-linked C telopeptide of type I collagen, parathyroid hormone (PTH), and sclerostin indicate resorptive processes while N-terminal propeptide of type I procollagen (PINP), osteocalcin (OC), and bone-specific alkaline phosphatase (BAP) indicate formative processes (Varela & Jolette, 2018). A multi-faceted approach to analysing skeletal health utilizing both imaging devices and biomarkers allows for a systematic and dynamic assessment of skeletal responses to exercise or physical training.

Exercise is often considered an osteogenic stimuli; however, inadequate rest between vigorous bouts, consistently elevated training volumes, and low energy availability may lead to chronic elevation of resorptive cues, and amplifying negative skeletal characteristics (O'Leary et al., 2020; Yanovich et al., 2015). This has been researched extensively in traditional athletes such as cyclists (Baker & Reiser, 2017; Sherk et al., 2014), runners (Beermann et al., 2020; Warden et al., 2021), and rowers (Lundy et al., 2022; Baker et al., 2020), with collegiate athletes suffering from 5.7 bone stress injuries at a rate of 100,000 per exposures (Hoenig et al., 2022). Military personnel also frequently conduct repetitive and arduous physical training with inadequate rest periods that may reduce bone density and quality (Armstrong et al., 2004; Hauschild et al., 2016). These individuals can suffer from high injury rates, with 5–10% of new Army, Navy, and Marine recruits developing a bone stress injury within the first 3 months of basic training, in large part due to lower baseline physical fitness levels prior to enlistment (Kardouni et al., 2021; Lee & Center AFHS, 2011; Waterman et al., 2016). Previous epidemiological research from Waterman and colleagues (Waterman et al., 2016) indicated that 77.5% of diagnosed stress injuries during military training and deployment occurred in junior enlisted members (<25 years old) spanning all U.S. military branches between 2009 and 2012. For instance, Armstrong et al., (2004), evaluated 31 U.S. Naval Academy midshipman aged 18–19 years old with

stress fractures compared to matched controls throughout introductory summer training. They found those that suffered from a fracture, lost more body mass and had reduced bone mineral content (BMC) and bone strength compared to non-fracture controls throughout the study duration. It would be beneficial to gain an understanding of training-related bone changes in those who have yet to start military training such as college-aged Reserve Officers Training Corps (ROTC) cadets, in an effort to reduce bone injury risk during service.

ROTC cadets can begin preliminary military training and studies during their collegiate career, with the option to commission as officers after graduation. Traditional ROTC training programmes often mimic exercises they will encounter during service, such as running, marching/rucking, push-ups/pull-ups, and crunches. If these programmes are completed at similarly high volumes and intensities, they could result in similar bone injury patterns. An additional factor to consider is the developmental age of cadets. Those between the ages of 18–25 can still experience skeletal geometric adaptations to training and are in a key period of bone mineral content accrual. Training programmes which negatively alter bone morphology and metabolism could have lasting detriments to bone injury risk. Key factors to consider include technical experience, training history, physical fitness, auxiliary exercise engagement, previous injury history, body composition, and nutritional factors (Jacobs et al., 2014; Scott et al., 2015). For instance, Scott et al., (2015) evaluated 195 cadets aged 18–33 years, throughout 16 months of traditional ROTC training, finding that lower Military Science class rank and higher physical training exposure were strong predictors of lower extremity injury. While Crombie et al., (2012) found that over 6 months, collegiate-level ROTC training did not inhibit an accumulation of body fat which is also predictor of injury in military personnel; potentially further increasing bone injury risk once enlisted. Traditional training programmes have been under scrutiny for increasing injury risk while providing lacklustre improvements in military physical readiness over the past decade, but the extent to which an updated ROTC training programme might alter bone and improve physical fitness is lacking (Nindl et al., 2015).

The primary aim of this study was to examine changes in bone density and quality, serum sclerostin and PTH responses, body composition, muscle strength, and aerobic capacity in ROTC cadets who engaged in an updated training programme utilizing a High-Intensity Interval Training-style (HIIT) as compared to physically active matched controls. We hypothesized ROTC cadets would exhibit improvements in all skeletal, body

composition, and performance metrics post 6 months compared to their baseline values and demonstrate similar or greater improvements than physically active matched controls.

Materials and methods

Subjects

Approval for this study was granted by *University of Oklahoma* Institutional Review Board (#8338 and 8600) and was in accordance with the Declaration of Helsinki and its amendments. Methods have been described elsewhere (Baker et al., 2021) but in brief, voluntary written informed consent was obtained from all subjects. ROTC inclusion criteria included males and females between 18 and 30 years old who were an active member of either the Army or Navy college ROTC programme for at least 1 year. Inclusion criteria for Controls were males and females who were physically active $\geq 3\times/\text{week}$ and were matched for sex, age (± 2 yrs), and body mass (± 2.5 kg) to an enrolled ROTC participant. Exclusion criteria for both ROTC and Controls consisted of having a reported history of musculoskeletal diseases, current or past smoking, metal implants, pregnancy, and secondary amenorrhoea, which was defined as having no menses for more than three consecutive months, without contraceptive use (Elliott-Sale et al., 2021).

Research design

This prospective study was split into three phases (Figure 1). Phase one occurred in late November, concluding the Fall semester eight-week ROTC training block. Phase two started in late January after a four-week winter break for college students and marked the beginning of a new Spring semester ROTC eight-week training block. Phase three ended the study in early April, marking the end of the Spring ROTC training block. Controls were tested ± 12 days to their ROTC match across the entire six-month study and maintained their activity levels.

Training and activity

Both Army and Navy ROTC cadets completed identical training protocols during both fall and spring semesters. The traditional ROTC training programme, consisting of long-distance runs, rucking and marching, pull-ups, push-ups, and curl-ups, was replaced with a HIIT programme, which aimed to incorporate

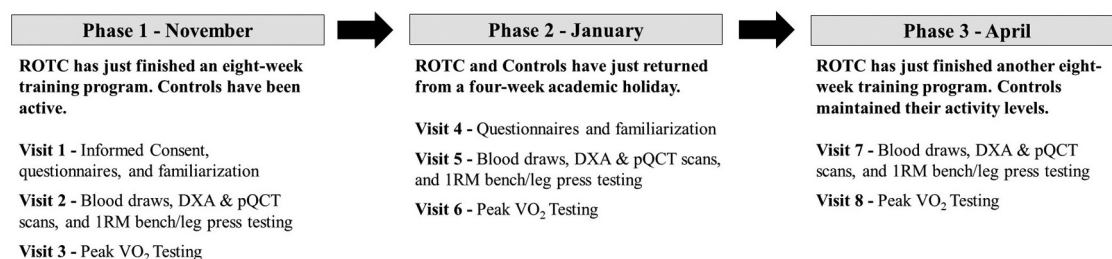


Figure 1. Phases and tests completed throughout the six-month evaluation period. Abbreviations include DXA: dual-energy X-ray Absorptiometry; pQCT: peripheral quantitative computed tomography; 1RM: 1 repetition maximum; peak VO₂: peak amount of oxygen consumption during the maximal aerobic capacity treadmill tests.

both resistance training and aerobic training within each session. All the exercise sessions (two sessions per week for 8 weeks, equating to 16 training hours) incorporated speed, agility, strength, and power movements. HIIT sessions and circuits were often followed by additional strength or endurance focused movements such as bench press, squat, runs, or cycling, providing a more comprehensive exercise programme. Control participants reported exercising $\geq 3\times/\text{week}$, with characterization of frequency and type indicated through self-reported training logs. Neither the ROTC or Controls exercise regimens were administered or altered by the research team in any way throughout the study.

Questionnaires

All subjects completed questionnaires regarding general health status, exercise and training logs, calcium intake (Musgrave et al., 1989), and bone-specific physical activity (BPAQ) (Weeks & Beck, 2008). Training logs included type, frequency, intensity, duration, and time of exercise. Questions specific to ROTC subjects included auxiliary exercise and physical activity outside of ROTC mandated training sessions and injury history such as stress fractures. The BPAQ was administered to quantify past, current, and total levels of osteogenic physical activity and has been shown to be positively correlated with bone geometry (Baker et al., 2020). All female subjects also completed an in-house menstrual history questionnaire to obtain information on cycle characteristics in the past 12 months such as contraceptive use, age at menarche, and symptoms of menstrual cycle and hormonal disturbances.

Bone and biomarkers

Subjects underwent DXA and pQCT scans for the assessment of body composition and bone measures (Baker et al., 2020, 2021). Urine samples were collected to ensure proper hydration 1.004–1.028 USG and confirmation of negative pregnancy status in females prior to scans. DXA (Lunar Prodigy, enCORE 16, GE Healthcare, Madison, WI) was used to measure total body fat mass (FM; g), % fat mass (BF%), bone-free lean body mass (BFLBM; g), and bone mineral content (BMC; g). Areal bone mineral density (aBMD; g/cm^2) was calculated for total body, lumbar spine (L1–L4), and proximal femora (femoral neck, greater trochanter, total hip) scans. Automated DXA data extraction was completed using the DXA Data Xtraction Assistant (DXA²) programme (Baker et al., 2021). Additionally, a pQCT scanner (XCT 3000, v.6.00, Stratec Medizintechnik GmbH, Pforzheim, Germany) was used to measure non-dominant 4%, 38%, and 66% tibial characteristics. At the distal tibia (4%), measures included total and trabecular measures of volumetric bone mineral density (vBMD; mg/cm^3), bone area (mm^2), periosteal circumference (mm), and estimated bone strength index (BSI; mg/mm^4). At the 38% and 66% tibia sites, variables included total and cortical measures of vBMD (mg/cm^3) and bone area (mm^2), cortical thickness (mm), polar moment of inertia (Ipolar; mm^4), periosteal and endosteal circumference (mm), and polar strength for strength-strain index (pSSI; mm^3). Muscle cross-sectional area (mCSA; mm^2) was also calculated for the 66% tibia site. All scans were visually rated as

a two or below and the average pMovement was 45.9 indicating scans with little to no movement as described by Blew et al., (2014). In our laboratory, the coefficient of variance percentage for all DXA and pQCT measurements ranges from 0.3% to 2.7% and the same qualified and trained technicians performed all quality assurance tests, scans, and analyses.

Enzyme-linked immunosorbent assays (ELISAs) for serum PTH and sclerostin were completed for all time points. Subjects were instructed to not exercise or eat for 24 and 8 hours, respectively, prior to blood draws. Approximately 10 mL were collected via venipuncture between 8:00 and 9:00 am. Clotted samples were centrifuged, aliquoted, and stored at -80°C . All samples were assayed in duplicate using DRG International Inc., Springfield, NJ. (Cat# EIA3645) for PTH and TECO medical Quidel Corp., Santa Clara, CA and Sissach, Switzerland (Cat# TE1023-HS) for sclerostin. For all assays, the intra- and inter-assay coefficient of variance percentage ranged from 0.2 to 9.4% and 4.7% to 8.5%, respectively.

Physical performance testing

All subjects completed upper and lower body assessments of maximal muscle strength and peak aerobic capacity tests. One repetition maximum (1RM) testing for bench press (Cybex, Medway, MA) and leg press (Body Solid, Forest Park, IL) followed National Strength and Conditioning Association recommendations (Miller, 2012); with interclass correlation coefficients for 1RM testing in our laboratory being above 0.990. Peak aerobic capacity (VO_2 peak) was measured using a modified Balke (Baker et al., 2021) treadmill protocol with open-circuit spirometry (ParvoMedics; Sandy, UT) and heart rate was monitored continuously using a coded transmitter worn around the chest (Polar T31, Bethpage, NY). Prior to the end of each stage, subjects were asked their Rating of Perceived Exertion (RPE) using the 15 point (6–20) Borg Scale (30). Four criteria were used to identify maximal oxygen uptake; a plateau in oxygen consumption despite an increased workload, Respiratory Exchange Ratio (RER) values over 1.10, RPE values 18 or over, and maximal Heart Rate (HR) within 10 bpm of the age predicted max HR. VO_2 peak was calculated as the average of the two highest consecutive 30-second VO_2 measurements.

Statistical analysis

An *a priori* power analysis using results from Janik et al., (2018), for serum sclerostin concentration, Vaara et al., (2015), for 1 RM bench press, and O'leary et al., (2021), for total body aBMD, were conducted. At 80% power, effect sizes were indicated as 0.6, 0.5, and 0.2 for each metric, respectively, suggesting a sample size of between 6 and 28 for 80% power. All statistical procedures were performed using IBM SPSS (v26, Armonk, New York), and significance was set at $p \leq 0.05$. All data were normally distributed based on the Shapiro–Wilks test and are reported as unadjusted means $\pm$ standard deviations (SD) in tables and means $\pm$ standard errors (SE) in figures. Baseline differences in physical characteristics between ROTC and Controls were examined using independent *t*-tests. A 2×3 repeated measures analysis of variance (RMANOVA) was used

to determine group (ROTC, Controls) and time (Nov, Jan, Apr) main effects and group by time interactions. Significant interactions were decomposed using one-way ANOVAs with Fisher's Least Significant Difference post-hocs for each time point.

Results

Participant descriptives

Initially, 36 subjects (ROTC $n = 18$, Controls $n = 18$) were screened for the study. Two ROTC were excluded prior to participation, one for prolonged illnesses and another withdrew after visit one, so their respective controls were also excluded. One more control was excluded after the winter break for not being able to maintain matching criteria. In total, 31 subjects (ROTC $n = 16$; Controls $n = 15$) were included in the final analysis. Females comprised 22% of the sample ($n = 8$) and 50% of females reported using oral contraceptives, while an additional 13% reported using an implant contraceptive method. No significant differences were found between ROTC and Controls' baseline data for all anthropometrics, questionnaire responses, calcium intake, exercise volumes, or BPAQ scores (all $p \geq 0.062$; Table 1). Calcium intake was not significantly different between groups ($p = 0.062$); however, ROTC were meeting the recommended daily intake (Insitutie of Medicine US, 2000), while Controls were not, so an exploratory analysis included it as a potential covariate. This analysis did not change any statistical outcomes, so only RMANOVA data are reported. Additionally, there were no significant differences between groups at baseline for bone (all $p \geq 0.111$; Tables 2 and 3) or biomarker and performance variables (all $p \geq 0.052$; Table 4).

Changes over time

Multiple significant group by time interactions and main effects for time were observed for DXA measures of the total hip aBMD and BMC (all $p \leq 0.050$; Table 2, Figure 2, Supplemental Digital Content - Table 1). Model decomposition via one-way ANOVAs and post hoc analyses revealed a significant increase in dominant total hip aBMD and non-dominant femoral neck BMC in ROTC from November to April (all $p \leq 0.030$). However, ROTC demonstrated a significant decrease in dominant and non-dominant greater trochanter from November to April (both p

Table 1. Participant characteristics at baseline testing, data are shown as mean (SD).

Measures	ROTC ($n = 16$)	Controls ($n = 15$)	Independent t-test p
Age (years)	20.4 (2.5)	21.2 (1.9)	0.445
Height (m)	1.76 (0.10)	1.79 (0.07)	0.325
Body Mass (kg)	74.4 (11.1)	76.8 (10.1)	0.543
Calcium Intake (mg/day)	1118 (668)	737 (378)	0.062
Total PA (days/wk)	5.0 (1.4)	5.0 (1.0)	0.996
Total RT (days/wk)	4.2 (1.5)	4.0 (1.4)	0.726
Total ET (days/wk)	3.3 (1.7)	2.3 (1.9)	0.130
BPAQ – Total	21.7 (14.4)	33.6 (22.0)	0.084
BPAQ – Past	38.3 (28.7)	62.8 (43.4)	0.072
BPAQ – Current	5.1 (1.7)	4.4 (2.6)	0.368

No significant differences were found between groups. Abbreviations: SD: Standard Deviation; 1RM: one repetition maximum; PA: Physical Activity; RT: Resistance Training; ET: Endurance Training, BPAQ: Bone-specific Physical Activity Questionnaire.

< 0.050). It should be noted that despite reaching statistical significance, this decrease did not exceed the least significant change of the DXA machine and should not be considered. Additionally, ROTC dominant total hip BMC significantly increased from November to April (post hoc $p = 0.047$).

Significant group by time interactions and main effects for time were observed for pQCT measures at all sites (all $p \leq 0.038$; Table 3, Supplemental Digital Content - Table 2). At the 4% site, significant interactions were found for total and trabecular vBMD and area and periosteal circumference (all $p \leq 0.034$); however, after post hoc analyses no significant group or time effects remained (all $p \geq 0.054$). At the 38% and 66% sites total vBMD and content, and cortical vBMD all significantly increased from November to April in ROTC (all $p \leq 0.029$). Lastly, 66% cortical area, thickness, and mCSA all increased from November to April in ROTC (all $p \leq 0.025$).

Sclerostin significantly increased from November to January in ROTC ($p < 0.001$) and remained elevated by April ($p = 0.278$), while PTH did not change in either group (Table 4 and Figure 3). When considering physical performance improvements (Table 4), leg press significantly increased during each testing phase similarly between groups (all post hoc $p \leq 0.040$), while bench press increased from November to April for both groups ($p = 0.013$). Peak VO_2 was significantly greater in ROTC compared to Controls at all time points (group effect $p = 0.020$) but both groups improved their aerobic capacity during each testing phase to a similar extent (all time main effect post hocs $p \leq 0.022$).

Table 2. DXA measures of body composition and bone over time, data are shown as mean (SD).

Measures	ROTC ($n = 16$)			Controls ($n = 15$)		
	November	January	April	November	January	April
Total Body Fat %	19.48 (6.01)	20.27 (5.51)	19.83 (5.81)	20.07 (5.38)	20.12 (5.72)	19.83 (5.54)
Lean Mass:Fat Mass	4.34 (1.52)	4.04 (1.26)	4.23 (1.52)	4.10 (1.31)	4.16 (1.49)	4.21 (1.52)
Total aBMD (g/cm^2)	1.34 (0.12)	1.34 (0.11)	1.33 (0.12)	1.36 (0.12)	1.37 (0.12)	1.36 (0.11)
L1-L4 aBMD (g/cm^2)	1.36 (0.21)	1.31 (0.12)	1.32 (0.12)	1.34 (0.17)	1.35 (0.17)	1.35 (0.15)
Dom Hip aBMD (g/cm^2)						
Femoral Neck	1.24 (0.14)	1.24 (0.14)	1.28 (0.15)	1.28 (0.13)	1.28 (0.14)	1.26 (0.14)
Greater Trochanter	1.00 (0.12)	0.99 (0.13) *	0.99 (0.12) ‡	1.00 (0.14)	0.99 (0.14) *	0.98 (0.14) ‡
Total Hip	1.22 (0.13)	1.22 (0.14) *	1.22 (0.13) ‡	1.24 (0.14)	1.23 (0.14) *	1.23 (0.14) ‡
ND Hip aBMD (g/cm^2)						
Femoral Neck	1.24 (0.13)	1.24 (0.13)	1.24 (0.13)	1.29 (0.13)	1.28 (0.14)	1.28 (0.14)
Greater Trochanter	1.00 (0.13)	0.99 (0.12)	0.99 (0.13) ‡	1.02 (0.15)	1.02 (0.17)	1.00 (0.15) ‡
Total Hip	1.22 (0.13)	1.22 (0.13)	1.22 (0.13)	1.24 (0.15)	1.23 (0.15)	1.23 (0.15)

Significance set at $p \leq 0.05$; * significant time effect from November to January; ‡ significant time effect from January to April; † significant time effect from November to April. Abbreviations: SD: Standard Deviation; aBMD: areal Bone Mineral Density; L1-L4: lumbar spine 1–4; Dom: dominant; ND: non-dominant.

Table 3. pQCT measures of bone and muscle over time, data are shown as mean (SD).

Measures	ROTC (n = 16)			Controls (n = 15)		
	November	January	April	November	January	April
4% Tibia						
Tot vBMD (mg/cm ³)	345.9 (29)	346.2 (27)	346.5 (27)	365.3 (36)	363.6 (37)	361.1 (38)
Tot Area (mm ²)	1175.6 (182)	1171.4 (180)	1170.7 (177)	1137.1 (206)	1149.0 (206)	1155.4 (209)
Tot BSI (mg ² /mm ⁴)	140.6 (27)	140.4 (27)	140.6 (27)	151.5 (34)	151.5 (34)	150.3 (35)
Trab vBMD (mg/cm ³)	310.2 (27)	310.5 (27)	310.8 (26)	317.4 (33)	317.1 (34)	315.4 (35)
Trab Area (mm ²)	1074.1 (177)	1071.9 (176)	1070.5 (171)	1020.6 (204)	1033.0 (205)	1041.5 (207)
Trab BSI (mg ² /mm ⁴)	103.8 (24)	103.9 (24)	103.9 (24)	103.5 (30)	104.4 (30)	104.2 (30)
38% Tibia						
Tot vBMD (mg/cm ³)	933.5 (57)	935.4 (58) *	937.1 (57) ‡	937.1 (67)	940.3 (69) *	939.6 (68) ‡
Tot Cont (mg/mm)	399.5 (55)	400.5 (55)	402.0 (56) ‡	425.6 (65)	425.9 (64)	426.0 (64) ‡
Cort vBMD (mg/cm ³)	1174.2 (26)	1175.8 (27) *	1176.3 (26) ‡	1162.0 (28)	1166.6 (29) *	1165.5 (28) ‡
Cort Cont (mg/mm)	384.7 (52)	385.5 (51)	407.9 (59)	407.5 (60)	407.8 (60)	407.9 (59)
Cort Thickness (mm)	6.0 (0.6)	6.0 (0.6)	6.0 (0.7)	6.3 (0.7)	6.3 (0.7)	6.3 (0.7)
EndoCirc (mm)	35.3 (5.2)	35.3 (5.2)	35.3 (5.2)	35.9 (7.3)	35.9 (7.2)	35.9 (7.2)
66% Tibia						
Tot vBMD (mg/cm ³)	694.5 (52)	716.9 (92)	698.5 (52) ‡	723.7 (71)	728.0 (71)	728.4 (71) ‡
Tot Cont (mg/mm)	439.6 (61)	440.6 (63) *	441.8 (63) ‡	464.8 (71)	466.3 (70) *	466.8 (70) ‡
Cort vBMD (mg/cm ³)	1134.4 (23)	1137.5 (27)	1136.9 (23) ‡	1130.4 (26)	1131.3 (24)	1133.6 (26) ‡
Cort Cont (mg/mm)	399.8 (56)	401.7 (55) *	402.4 (58) ‡	420.1 (61)	451.6 (61) *	422.7 (61) ‡
Cort Thickness (mm)	4.7 (0.5)	4.9 (0.5)	4.8 (0.5) ‡	5.0 (0.6)	5.0 (0.6)	5.1 (0.6) ‡
EndoCirc (mm)	59.3 (6.9)	57.6 (10.5)	59.1 (6.7) ‡	58.3 (8.5)	58.1 (8.5)	58.0 (8.4) ‡
mCSA (mm ²)	7874.9 (1005.9)	7716.4 (928.3)	7931.5 (920.8) †	8332.4 (1379.6)	8417.6 (1512.3)	8471.8 (1611.7) †

Significance set at $p \leq 0.05$; * significant time effect from November to January; † significant time effect from January to April; ‡ significant time effect from November to April. Abbreviations: SD: Standard Deviation; Tot: Total; vBMD: volumetric Bone Mineral Density; Trab: Trabecular; BSI: Bone Strength Indices; Cort: Cortical; Endo Circ: Endosteal Circumference; mCSA: muscle Cross-Sectional Area.

Table 4. Changes in serum biomarkers and performance measures over time, data are shown as mean (SD).

Measures	ROTC (n = 16)			Controls (n = 15)		
	November	January	April	November	January	April
Biomarkers						
Sclerostin (ng/mL)	0.38 (0.09)	0.44 (0.14) *	0.42 (0.12) ‡	0.37 (0.09)	0.42 (0.09) *	0.14 (0.11) ‡
Parathyroid Hormone (U/L)	45.95 (12.47)	39.03 (20.87)	43.98 (21.68)	40.68 (12.27)	41.90 (14.43)	42.95 (16.12)
Performance						
1RM Leg Press (kg)	242 (83)	266 (75) *	292 (82) † ‡	280 (89)	289 (89) *	315 (97) † ‡
1RM Bench Press (kg)	83 (29)	84 (29)	87 (29) ‡	87 (33)	87 (32)	89 (32) ‡
VO ₂ Peak (ml/kg/min)	51.4 (7.1) #	53.3 (7.9) *#	54.3 (8.4) † ‡ #	46.8 (5.4)	47.7 (5.0) *	49.4 (5.6) † ‡
Respiratory Exchange Ratio	1.21 (0.06)	1.15 (0.06)	1.13 (0.07)	1.18 (0.06)	1.17 (0.07)	1.12 (0.04)
Maximal Heart Rate (bpm)	199 (6)	197 (6)	196 (6)	199 (7)	195 (6)	196 (8)
Rate of Perceived Exertion	17 (1)	18 (1)	18 (1)	18 (1)	18 (1)	19 (1)

Significance set at $p \leq 0.05$; * significant time effect from November to January; † significant time effect from January to April; ‡ significant time effect from November to April; # significant group effect. Abbreviations: SD: Standard Deviation; 1RM: one repetition maximum.

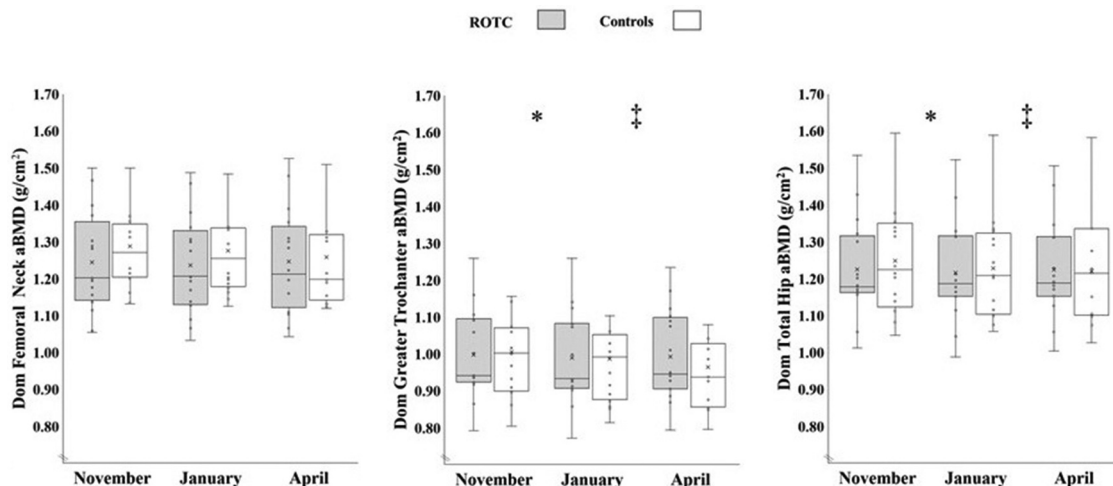


Figure 2. Dominant femoral neck, greater trochanter, and total hip aBMD throughout the six-month evaluation period are shown, from left to right respectively. ROTC data is in the grey bars while control data is in the white bars. Greater trochanter and total hip aBMD significantly increased from November to January (* both $p \leq 0.009$) and remained significantly elevated in April (‡ both $p \leq 0.016$) for both groups. Abbreviations included Dom: dominant limb; aBMD: areal bone mineral density.

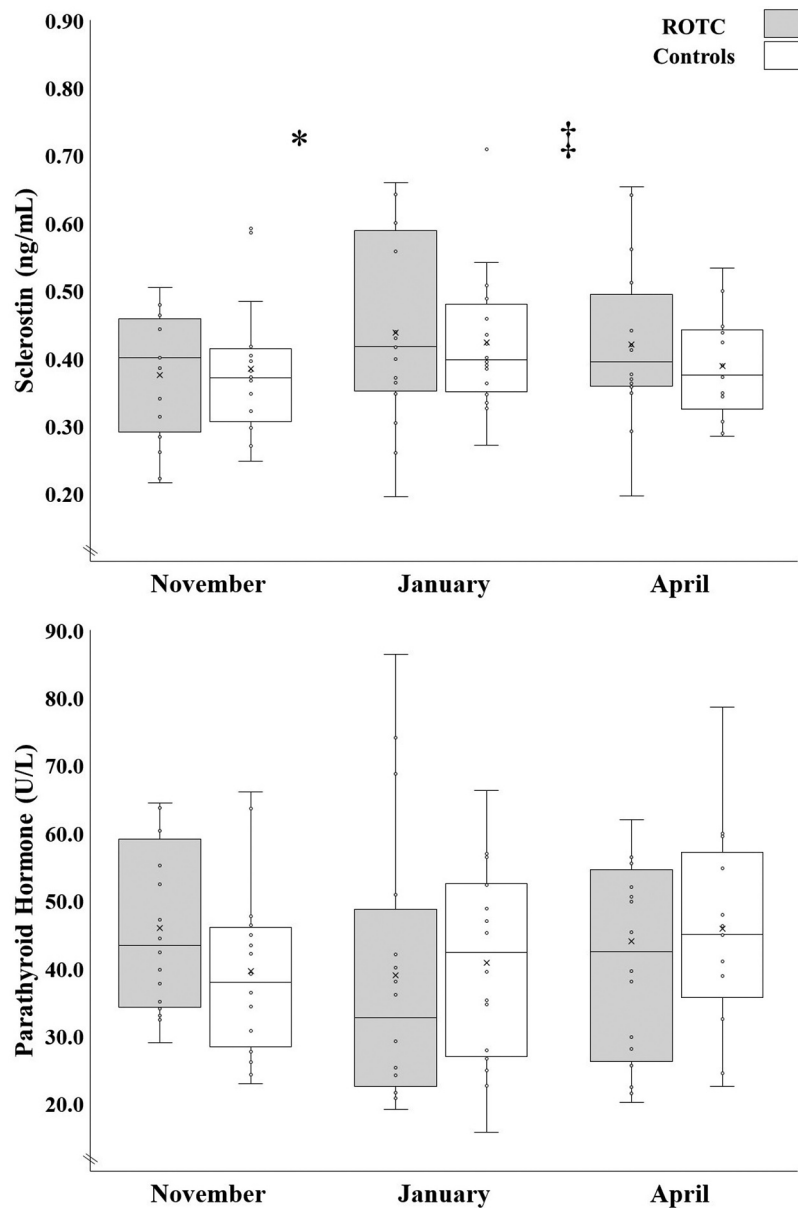


Figure 3. Serum concentrations of sclerostin and PTH throughout the six-month evaluation period are shown in the top and bottom panels, respectively. Sclerostin significantly increased from November to January (* $p < 0.001$) and remained significantly elevated in April (‡ $p = 0.007$). PTH did not change over time ($p = 0.342$) and no group effects were found for either biomarker ($p = 0.744$). ROTC data is in the grey bars while control data is in the white bars. Exclusion of outliers (greater than two standard deviations) did not change statistical findings and thus all the data are included.

Discussion

Due to the physically arduous nature of military training and high injury incidence, longitudinal evaluation of musculoskeletal health and performance is critical (Armstrong et al., 2004; Popp et al., 2021). This study temporally evaluated skeletal health, biomarker responses, and physical performance changes in college-aged ROTC cadets over 6 months, finding an updated ROTC training programme based on HIIT principles elicited positive musculoskeletal adaptations which, if maintained, would be skeletally protective during their future military training.

Just 5 years ago, overuse injuries accounted for nearly 70% of all reported musculoskeletal injuries among active-duty Army personnel (Molloy et al., 2020), costing over \$115 million (Forrest et al., 2021). These bone stress injuries are often

characterized as either mineral insufficiency or fatigue-derived overuse fractures and the risk of either type may be reduced by implementing the HIIT style training. First, the new programme effectively deviates from traditional endurance-based programmes (Schofield & Kardouni, 2015) by including many functional, multi-vector loading-based circuits. The improvements in ROTC bone density and content of the total hip and bone geometry at both 38% and 66% tibial sites most likely benefited from the introduction of these multi-directional loading patterns, effectively increasing load resistance and reducing insufficient fracture risk. Our results support those of Hughes et al., (2018) and O'Leary et al., (2019) who both assessed bone quality and strength in 91 female U.S. Army recruits and 43 male British Army recruits, respectively. Both studies noted positive tibial bone geometry adaptations similar to the current study and noted

these improvements were most likely attributed to participants' exposure to novel loading patterns. These improvements in ROTC may be especially important due to the developmental age of the cadets as positive bone adaptations now can have lasting benefit to the skeleton and potentially reduce future injury risk.

Fatigue fracture risk may also be reduced in new ROTC training programmes due to the adoption of HIIT foundations. During HIIT training, ROTC may be able to see performance improvements that meet or exceed those of a traditional programme, while engaging in reduced training volumes. For instance, Gist et al., (2015), compared the efficacy of 4 weeks of HIIT training versus the traditional ROTC training programme in 26 college-aged cadets. HIIT training required less than half of the same training time and volume compared to the traditional training, but Army Physical Fitness Test performance was similar. Using HIIT reduced the overall training volume, shifted the loading profile from repetitive to novel, and increased the rest periods, all of which can be substantial stimuli for positive bone growth. HIIT has also been shown to improve bone more than traditional training programmes in adolescents (Julian et al., 2022) and college-aged adults (Lu et al., 2022). Taken together, it appears that this updated training programme which utilized HIIT as the foundation has the ability to reduce risk of both insufficiency and fatigue fractures by improving bone content, density, and geometry while also reducing training volumes and increasing rest periods.

The updated ROTC programme also positively impacted many factors that are indirectly associated with bone injuries. For instance, Beck et al., (2000) reported military recruits suffering from stress injuries demonstrated poorer fitness levels, smaller mid-thigh mCSA, and reduced estimated thigh and tibiae bone strength compared to their injury-free counterparts. These findings were corroborated by Hoffman et al., (1999) who found 18-year-old army recruits with lower extremity strength greater than one standard deviation below the population mean were at a significantly greater estimated risk for stress fractures. Beck et al., (2000) hypothesized that those with reduced muscle strength and size would not be able to adequately dissipate ground reaction forces during training, leading to increased localized skeletal loading, and the accumulation of microtrauma resulting in bone injury. Our findings support those of Beck et al., (2000) and Hoffman et al., (1999) as ROTC from the current study significantly improved their muscle strength and mCSA, which coincided with improved bone geometry. Furthermore, Armstrong et al., (2004) evaluated bone health using DXA in 31 Naval Academy cadets aged 18.7 years over the summer months. They found cadets who suffered a stress fracture had greater reductions in body mass and BMC compared to those who did not fracture throughout a traditional training regimen, suggesting weight maintenance to be a key factor. In the current study, ROTC participants maintained body mass and BMC while reporting no fractures during the intervention period. Taken together, these data suggest that the newly adapted ROTC training programme also improved many indirect correlates of bone injury, potentially further reducing future risk.

The biomarkers sclerostin and PTH were included in this study to provide a metabolic indicator of bone metabolism over time.

PTH did not change, and the literature surrounding this biomarker's response to military training is mixed as previously noted. However, sclerostin significantly increased during the winter break and remained elevated for the remainder of the study, posing some interesting questions. As sclerostin is a potent inhibitor of bone formation, high serum concentrations are often linked with periods of physical inactivity. At first, our data confirmed this relationship as sclerostin increased during the winter break, when training was not mandated. However, activity levels as estimated via BPAQ show no difference in activity from November to January suggesting a maintenance in activity. Additionally, these high serum sclerostin values remained elevated during the second eight-week training period from January to April. These data suggest high rates of bone resorption; however, bone loss was not observed from DXA or pQCT measures. We hypothesize that during the intervention period participants were indeed experiencing increased rates of bone resorption as evidenced by the elevated sclerostin concentrations; however, bone formation outpaced this catabolic activity to result in the observed increases in bone mineral deposition. Without definitive assessments of bone formation from biomarkers, such as PINP, OC, or BAP, this is purely speculative.

This study presents several strengths and limitations to consider. First, this study assessed ROTC's bone health and performance using DXA, pQCT, serum biomarkers, 1RM, and peak VO₂ testing across multiple training interventions. And while these data are valid, reliable, and precise, they are not tools that can always be easily utilized in the field with hundreds of young military personnel engaging in basic training blocks, reducing the potential generalizability of these data. Second, as previously noted, measures of both bone resorption and formation must be included in future research to have a better understanding of the magnitude of metabolic shifts that may occur in response to training. Third, although physical activity frequency was collected, resistance and endurance training intensity were not recorded which may influence the magnitude of musculoskeletal adaptation to training and should serve as a pertinent measure for future investigations. Finally, our cohort was comprised almost entirely of male cadets and future research needs to consist of a more diverse cohort, especially since women now make up a larger proportion of enlisted military than ever before.

In conclusion, a non-traditional HIIT-style ROTC training regimen elicited positive skeletal adaptations which coincided with improved muscle size and strength. These findings indicate collegiate ROTC cadets should perhaps engage in more HIIT-style training programmes prior to service to reduce the risk of bone injury.

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